TIØ4145 Corporate Finance Chapter 17

Maria Lavrutich
Email: maria.lavrutich@iot.ntnu.no
Tel: 735 93189
Web: www.ntnu.no/ansatte/maria.lavrutich

Department of Industrial Economics and Technology Management
Norwegian University of Science and Technology
Monday, October 6, 2025

NTNU
Norwegian University of
Science and Technology

Chapter 17: Payout Policy

17.1 Distributions to Shareholders

17.2 Comparison of Dividends and Share Repurchases

17.3 The Tax Disadvantage of Dividends

17.4 Dividend Capture and Tax Clienteles

17.5 Payout Versus Retention of Cash

17.6 Signaling with Payout Policy

17.7 Stock Dividends, Splits and Spin-offs

NTNU
Norwegian University of
Science and Technology

Learning Objectives

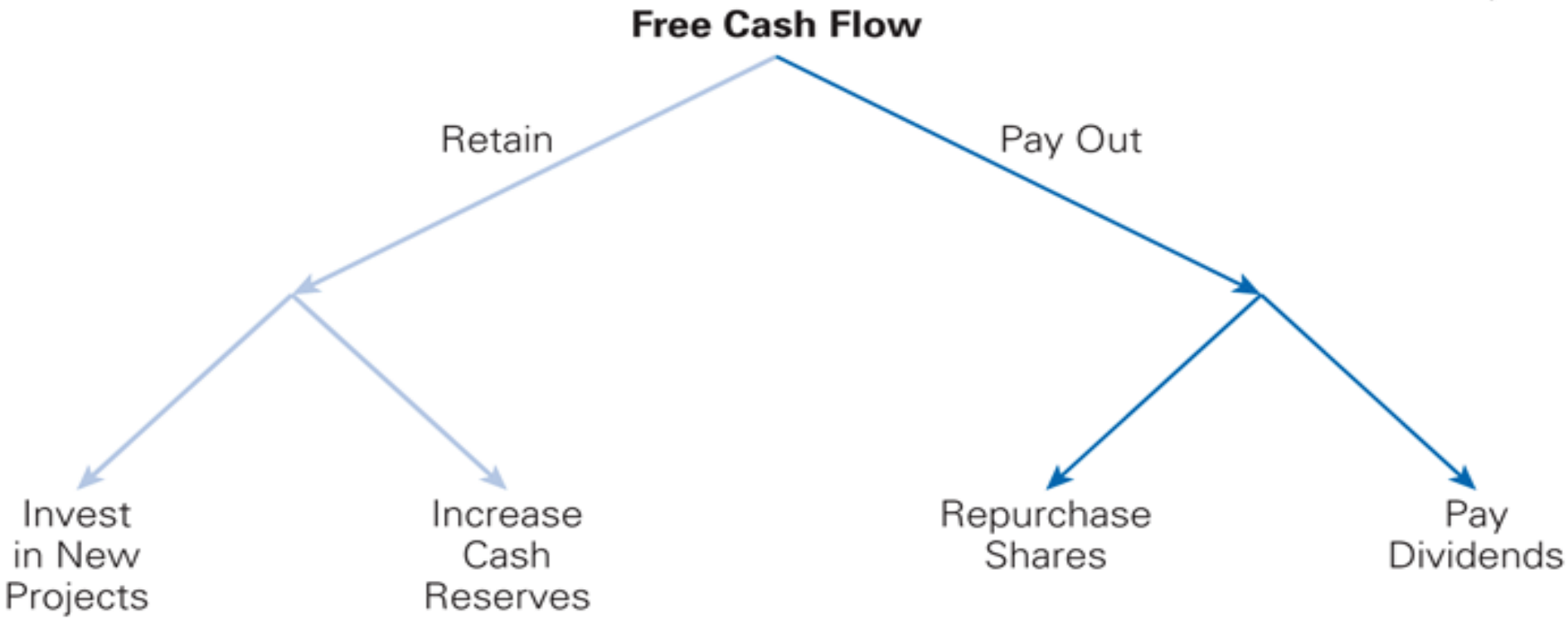
- Discuss the effect of dividend payment or share repurchase in perfect capital markets.
- Assuming perfect capital markets, describe what Modigliani and Miller (1961) found about payout policy.
- Describe th effect of tax on payout policy
- Provide reasons why firms might accumulate cash balances rather than pay dividends.
- Describe the effect of agency costs on payout policy.
- Assess the impact of information asymmetry on payout policy.

NTNU
Norwegian University of
Science and Technology

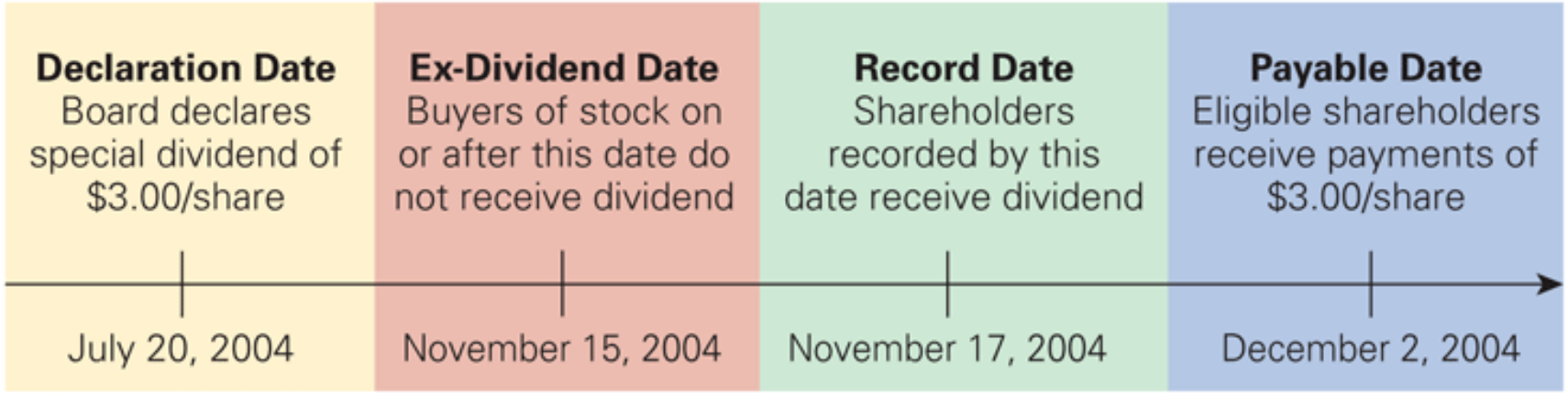
Distribution to Shareholders

Payout Policy

- The way a firm chooses between the alternative ways to distribute free cash flow to equity holders



Dividends



Declaration Date

- The date on which the board of directors authorizes the payment of a dividend

Ex-dividend Date

- A date, two days prior to a dividend's record date, on or after which anyone buying the stock will not be eligible for the dividend

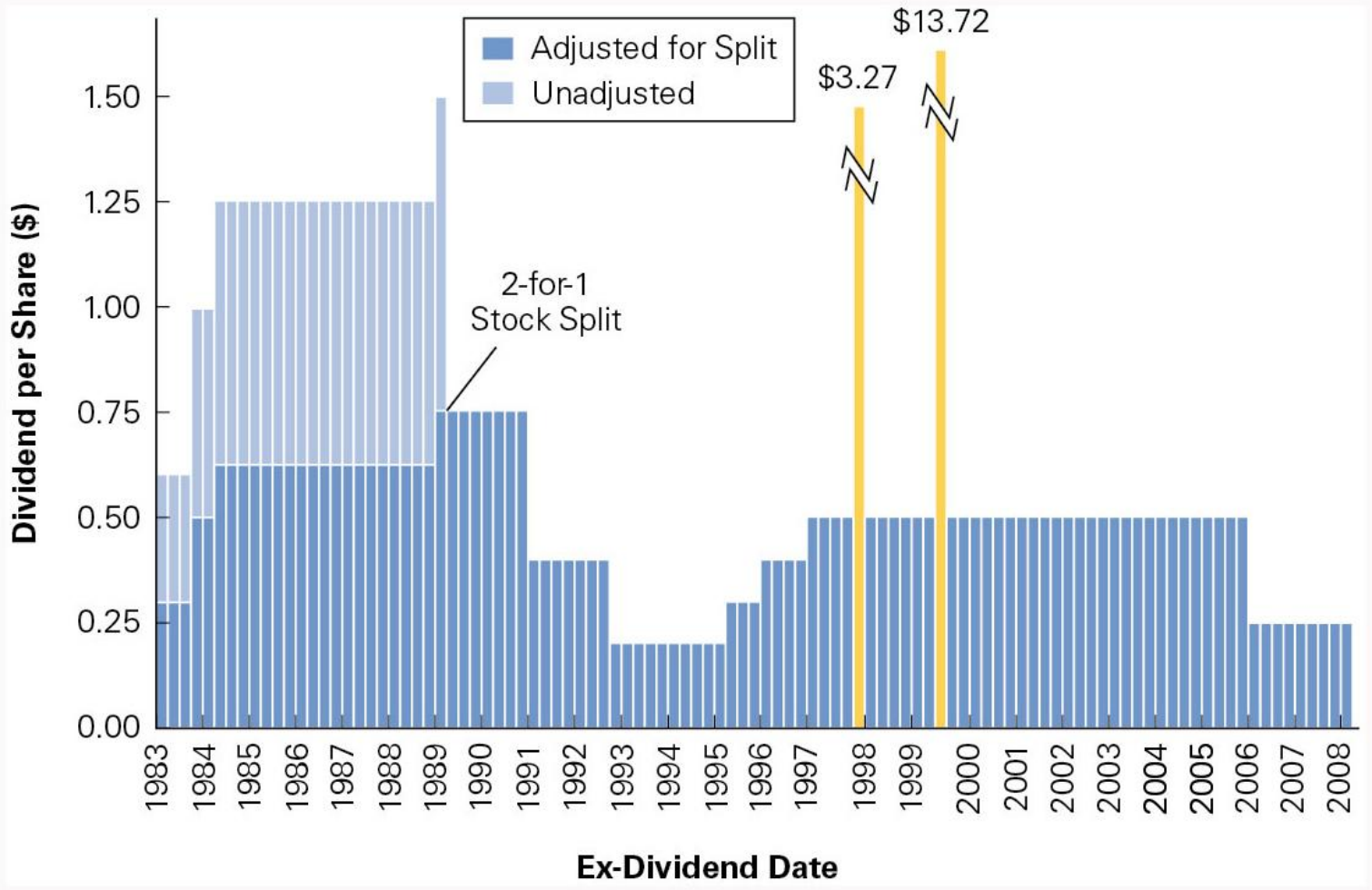
Record Date

- When a firm pays a dividend, only shareholders on record on this date receive the dividend.

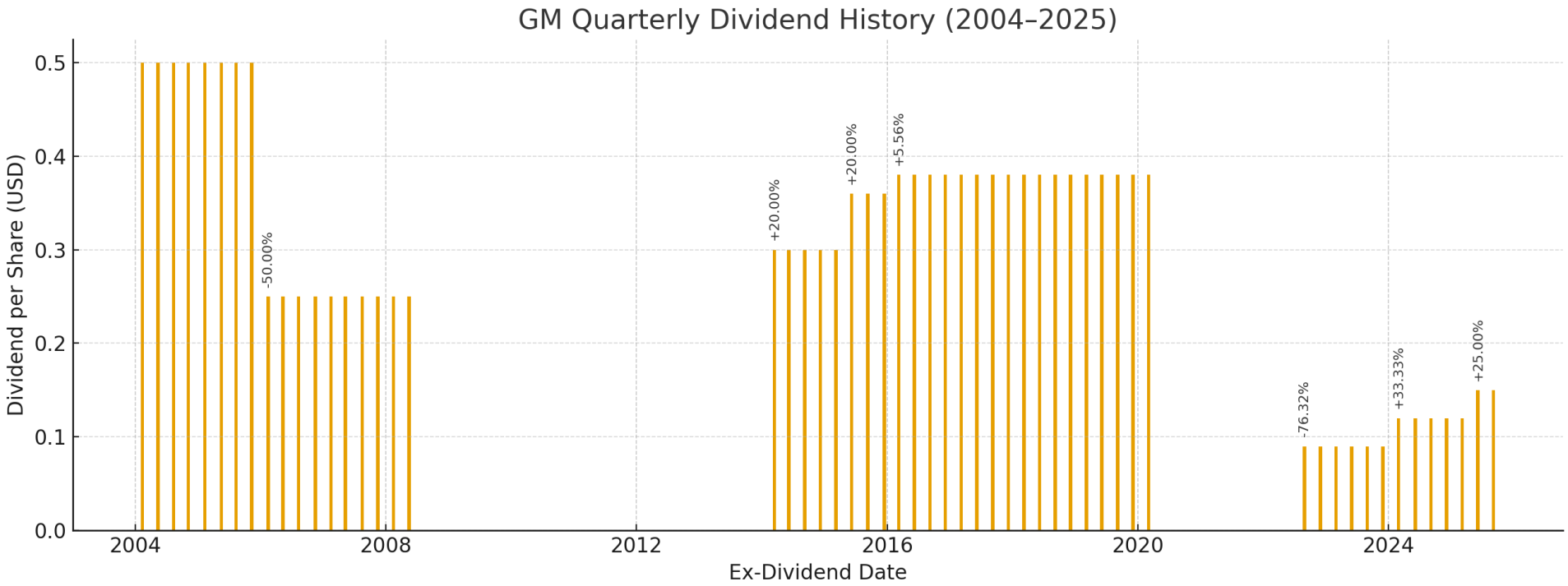
Payable Date (Distribution Date)

- A date, generally within a month after the record date, on which dividend is paid

Figure 17.3 Dividend History for GM Stock, 1983–2008



Dividend History for GM Stock, 2008–2025



Source: https://www.digrin.com/stocks/detail/GM/

NTNU
Norwegian University of Science and Technology

Share Repurchases

- An alternative way to pay cash to investors is through a share repurchase or buyback.
  - The firm uses cash to buy shares of its own outstanding stock.
- There are different transaction types for share repurchases (see Chapter 17.1)

NTNU
Norwegian University of
Science and Technology

Comparison of Dividends and Share Repurchases

Example of Genron Corporation in Section 17.2:

- The firm's board is meeting to decide how to pay out $20 million in excess cash to shareholders.
- Genron has no debt, its equity cost of capital equals its unlevered cost of capital of 12%.
- Genron expects to generate future free cash flows of $48 million per year in subsequent years.
- Genron has 10 million outstanding shares.

NTNU
Norwegian University of Science and Technology

Comparison of Dividends and Share Repurchases

Compare different potential policies:

- Pay dividend with excess cash
- Share repurchase (no dividend)
- High Dividend (equity issue)

Questions:

- Which policy would shareholders prefer?
- Will the amount of the dividend affect Genron's share price?

NTNU
Norwegian University of Science and Technology

Alternative Policy 1: Pay Dividend with Excess Cash

- With 10 million shares outstanding, Genron will be able to pay a $2 dividend immediately.
- The firm expects to generate future free cash flows of $48 million per year, thus it anticipates paying a dividend of $4.80 per share each year thereafter.
- The price of Genron will be

$$P_{cum} = \text{Current Dividend} + PV(\text{Future Dividends}) = 2 + \frac{4.80}{0.12} = 2 + 40 = \$42$$

NTNU
Norwegian University of
Science and Technology

Alternative Policy 1: Pay Dividend with Excess Cash

- After the ex-dividend date, new buyers will not receive the current dividend and the share price and the price of Genron will be

$$P_{ex} = PV(\text{Future Dividends}) = \frac{4.80}{0.12} = \$40$$

	December 11 (Cum-Dividend)	December 12 (Ex-Dividend)
Cash	20	0
Other assets	400	400
Total market value	420	400
Shares (millions)	10	10
Share price	**$42**	**$40**

NTNU
Norwegian University of Science and Technology

Alternative Policy 1: Pay Dividend with Excess Cash

- In a perfect capital market, when a dividend is paid, the share price drops by the amount of the dividend when the stock begins to trade ex-dividend.

Alternative Policy 2: Share Repurchase (No Dividend)

- Suppose that instead of paying a dividend this year, Genron uses the $20 million to repurchase its shares on the open market.
  - With an initial share price of $42, Genron will repurchase 476,000 shares.
    - $20 million ÷ $42 per share = 0.476 million shares
  - This will leave only 9.524 million shares outstanding.
  - 10 million − 0.476 million = 9.524 million

NTNU
Norwegian University of Science and Technology

Alternative Policy 2: Share Repurchase (No Dividend)

- The net effect is that the share price remains unchanged.

	December 11 (Before Repurchase)	December 12 (After Repurchase)
Cash	20	0
Other assets	400	400
Total market value of assets	420	400
Shares (millions)	10	9.524
Share price	**$42**	**$42**

Alternative Policy 2: Share Repurchase (No Dividend)

- Genron's Future Dividends
  - It should not be surprising that the repurchase had not effect on the stock price.
  - After the repurchase, the future dividend would rise to $5.04 per share.
    - $48 million ÷ 9.524 million shares = $5.04 per share
    - Genron's share price is

$$P_{rep} = \frac{5.04}{0.12} = \$42$$

In perfect capital markets, an open market share repurchase has no effect on the stock price

NTNU
Norwegian University of Science and Technology

Alternative Policy 1 compared to 2:

- Investor Preferences
  - In perfect capital markets, investors are indifferent between the firm distributing funds via dividends or share repurchases. By reinvesting dividends or selling shares, they can replicate either payout method on their own.

NTNU
Norwegian University of
Science and Technology

Alternative Policy 3: High Dividend (Equity Issue)

- Suppose Genron wants to pay dividend larger than $2 per share right now, but it only has $20 million in cash today. It wants to pay out $48 million
  - Thus, Genron needs an additional $28 million to pay the larger dividend now.
  - To do this, the firm decides to raise the cash by selling new shares.

NTNU
Norwegian University of Science and Technology

Alternative Policy 3: High Dividend (Equity Issue)

- Given a current share price of $42, Genron could raise $28 million by selling 0.67 million shares.
  - $28 million ÷ $42 per share = 0.67 million shares
    - This will increase the total number of shares to 10.67 million.

NTNU
Norwegian University of Science and Technology

Alternative Policy 3: High Dividend (Equity Issue)

- The new dividend per share will be

$$\frac{\$48 \text{ million}}{10.67 \text{ million shares}} = \$4.50 \text{ per share}$$

- And the cum-dividend share price will be

$$P_{cum} = 4.50 + \frac{4.50}{0.12} = 4.50 + 37.50 = \$42$$

- Again, the share value is unchanged.

NTNU
Norwegian University of
Science and Technology

Genron's Dividends per Share Each Year Under the Three Alternative Policies

	Initial Share Price	Dividend Paid ($ per share)			
		Year 0	Year 1	Year 2	. . .
Policy 1:	$42.00	2.00	4.80	4.80	. . .
Policy 2:	$42.00	0	5.04	5.04	. . .
Policy 3:	$42.00	4.50	4.50	4.50	. . .

Table 17.1

NTNU
Norwegian University of
Science and Technology

Modigliani–Miller and Dividend Policy Irrelevance

- There is a trade-off between current and future dividends.
  - If Genron pays a higher current dividend, future dividends will be lower.
  - If Genron pays a lower current dividend, future dividends will be higher.

- MM Dividend Irrelevance
  - In perfect capital markets, holding fixed the investment policy of a firm, the firm's choice of dividend policy is irrelevant and does not affect the initial share price.

NTNU
Norwegian University of Science and Technology

17.3 The Tax Disadvantage of Dividends and 17.4 Tax Clienteles

- Taxes on Dividends and Capital Gains
  - Shareholders must pay taxes on the dividends they receive and they must also pay capital gains taxes when they sell their shares.
  - In some countries dividends are taxed at a higher rate than capital gains. In that case shareholders might prefer share repurchases to dividends.
  - In Norway, the tax system is symmetric, and hence there is no such incentive

NTNU
Norwegian University of
Science and Technology

17.5 Payout Versus Retention of Cash

- MM Payout Irrelevance
  - In perfect capital markets, if a firm invests excess cash flows in financial securities, the firm's choice of payout versus retention is irrelevant and does not affect the initial share price.
- With market imperfections, there is a tradeoff: Retaining cash can reduce the costs of raising capital in the future, but it can also increase taxes and agency costs.

NTNU
Norwegian University of Science and Technology

Example

- Payne Enterprises has $20,000,000 in excess cash.
- Payne is considering investing the cash in one-year Treasury bill paying 5% interest and then using the cash to pay a dividend next year.
  - If Payne pays an immediate dividend, the shareholders receive $20,000,000 today.
  - If Payne retains the cash, at the end of one year the company will be able to pay a dividend of $20,000,000 × (1.05) = $21,000,000
  - If shareholders invest the $20,000,000 in Treasury bills themselves, they would $20,000,000 × (1.05) = $21,000,000
  - The present value in either scenario is $20,000,000
- Thus shareholders are indifferent about whether the firm pays the dividend immediately or retains the cash.

NTNU
Norwegian University of
Science and Technology

Market imperfections

- **Taxes**
  - Corporate taxes make it costly for a firm to retain excess cash.
  - Cash is equivalent to negative leverage, so the tax advantage of leverage implies a tax disadvantage to holding cash
- **Issuance and Distress Costs**
  - Generally, firms retain cash balances to cover potential future cash shortfalls, despite the tax disadvantage to retaining cash.
  - A firm might accumulate a large cash balance if there is a reasonable chance that future earnings will be insufficient to fund future positive-NPV investment opportunities.
  - Firms should choose to retain to help with future growth opportunities and to avoid financial distress costs.
- **Agency Costs**
  - When firms have excessive cash, managers may use the funds inefficiently by paying excessive executive perks, over-paying for acquisitions, etc.
  - Paying out excess cash through dividends or share repurchases, rather than retaining cash, can boost the stock price by reducing managers' ability and temptation to waste resources.

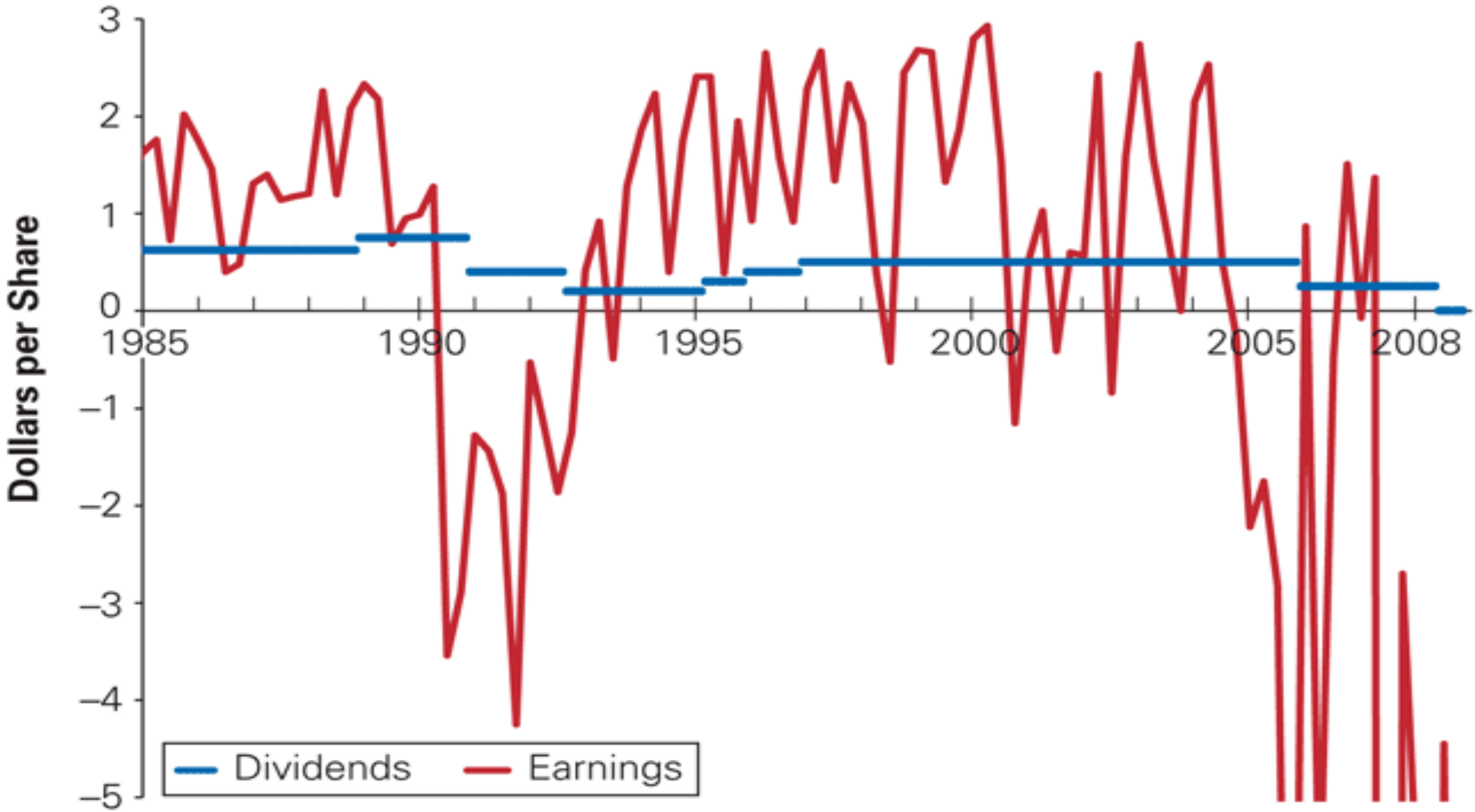
NTNU
Norwegian University of
Science and Technology

17.6 Signaling with Payout Policy

- The idea that dividend changes reflect managers' views about a firm's future earning prospects.
- Dividend Smoothing
  - The practice of maintaining relatively constant dividends
    - Firm change dividends infrequently and dividends are much less volatile than earnings
  - Research has found that management believes that investors desire to maintain a long-term target level of dividends as a fraction of earnings.
- If firms smooth dividends, the firm's dividend choice will contain information regarding management's expectations of future earnings.

NTNU
Norwegian University of Science and Technology

GM's Earnings and Dividends per Share, 1985–2008



Signaling and Share Repurchases

- Share repurchases are a credible signal that the shares are underpriced, because if they are overpriced a share repurchase is costly for current shareholders.
  - If investors believe that managers have better information regarding the firm's prospects and act on behalf of current shareholders, then investors will react favorably to share repurchase announcements.

NTNU
Norwegian University of Science and Technology

Practitioner's view

The company considered five key questions in developing its distribution strategy:

1. Can the company sustain payment of a cash dividend in perpetuity and increase the dividend over time?
2. Is a cash dividend a better return to stockholders than a stock buyback program? These are capital structure decisions: Do we want to reduce our shares outstanding? Is our stock attractively priced for a buyback, or do we want to distribute the cash as a dividend? Microsoft had plenty of capacity to issue a dividend *and* continue a buyback program.
3. What is the tax effect of a cash dividend versus a buyback to the corporation and to shareholders?
4. What is the psychological impact on investors, and how does it fit the story of the stock for investors? This is a more qualitative factor.
5. What are the public relations implications of a dividend program? Investors don't look to Microsoft to hold cash but to be a leader in software development and provide equity growth. So they viewed the dividend program favorably.

From the interview with John Connors former CFO at Microsoft (you can find it in the book).

NTNU
Norwegian University of
Science and Technology

Key take-aways Chapter 17

- In perfect capital markets, holding fixed the investment policy of a firm, the firm's choice of dividend policy is irrelevant and does not affect the initial share price.
- In perfect capital markets, if a firm invests excess cash flows in financial securities, the firm's choice of payout versus retention is irrelevant and does not affect the initial share price.
- With market imperfections, there is a tradeoff: Retaining cash can reduce the costs of raising capital in the future, but it can also increase taxes and agency costs.
- Dividends and share repurchases help minimize the agency problem of wasteful spending when a firm has excess cash.
- Dividend signaling hypothesis.
  - Managers usually increase dividends only when they are confident the firm will be able to afford higher dividends for the foreseeable future.
  - When managers cut the dividend, it may signal that they have lost hope that earnings will improve.
  - Share repurchases may be used to signal positive information, as repurchases are more attractive if management believes the stock is undervalued at its current price.

NTNU
Norwegian University of
Science and Technology